

Machine Learning for Electrocatalyst Discovery in Energy Conversion: A Comprehensive Survey

Abstract

The urgent transition toward sustainable energy systems necessitates the rapid discovery of efficient, stable, and cost-effective electrocatalysts for critical reactions such as the hydrogen evolution reaction (HER), oxygen evolution reaction (OER), oxygen reduction reaction (ORR), and carbon dioxide reduction reaction (CO₂RR). Traditional discovery paradigms, reliant on intuition-driven experimentation and computationally expensive density functional theory (DFT) calculations, are increasingly insufficient to navigate the vast chemical and structural space of potential materials. This survey provides a comprehensive review of the integration of machine learning (ML) and artificial intelligence (AI) into electrocatalyst discovery. We synthesize evidence from 300 recent academic studies to construct a detailed taxonomy of methodological advances, ranging from classical descriptor-based regression and symbolic regression to deep generative models, graph neural networks (GNNs), and large language model (LLM)-driven autonomous agents. We examine the evolution of data curation, highlighting the shift from small, curated datasets to massive, high-fidelity repositories like the Open Catalyst Project (OC20/OC22/OC25) and grand-canonical databases. Furthermore, we analyze the emergence of closed-loop self-driving laboratories that integrate robotic synthesis, characterization, and ML-guided active learning. Finally, we discuss persistent challenges regarding model generalizability, the accuracy-computational cost trade-off in machine-learned interatomic potentials, the necessity of uncertainty quantification, and the critical gap between theoretical prediction and experimental validation.

1. Introduction

Electrocatalysis lies at the heart of renewable energy conversion technologies, including water electrolyzers, fuel cells, and metal-air batteries. The efficiency of these devices is fundamentally limited by the kinetics of electrode reactions, which require catalysts to lower activation barriers and enhance selectivity. Historically, the development of electrocatalysts has been guided by the Sabatier principle and scaling relations, often leading to the dominance of precious metals like platinum, iridium, and ruthenium [1], [2]. However, the scarcity and high cost of these materials necessitate the discovery of earth-abundant alternatives with comparable or superior performance.

The chemical space for potential electrocatalysts is astronomically large, encompassing diverse compositions (alloys, oxides, sulfides, single-atom catalysts), morphologies (nanoparticles, nanosheets, porous structures), and operational conditions (pH, potential, temperature). Traditional high-throughput screening using DFT, while accurate, is computationally prohibitive for exploring this vast space exhaustively, often limiting studies to hundreds of candidates [3], [4]. Moreover, the “materials gap” between idealized theoretical models and complex experimental realities—such as surface reconstruction, solvation effects, and dynamic active site evolution—further complicates the design process [5], [6].

Machine learning has emerged as a transformative tool to accelerate this discovery pipeline. By learning complex, non-linear mappings between material descriptors and catalytic properties from existing data, ML models can predict performance orders of magnitude faster than DFT, enabling the screening of millions of candidates [7], [8]. The field has evolved from simple linear regression on hand-crafted descriptors to sophisticated deep learning architectures capable of processing raw atomic coordinates, text-based literature, and even generating novel crystal structures [9], [10],

[11]. Recent advancements include the deployment of multi-agent LLM systems for hypothesis generation and the integration of ML into autonomous robotic laboratories that close the loop between prediction, synthesis, and characterization [12], [13], [14].

This survey synthesizes the current state of ML for electrocatalyst discovery, drawing upon 300 distinct studies published between 2016 and 2026. We organize the discussion into a coherent methodological taxonomy, covering data infrastructure, descriptor engineering, predictive modeling, generative design, and autonomous discovery frameworks. We critically evaluate the strengths and limitations of each approach, emphasizing the interplay between computational efficiency, physical interpretability, and experimental fidelity.

2. Method

The application of machine learning in electrocatalysis can be categorized into several interconnected methodological families: data curation and representation learning, property prediction via supervised and symbolic regression, generative design for inverse materials discovery, and autonomous closed-loop discovery systems. This section details these approaches, synthesizing findings from the provided literature.

2.1 Data Infrastructure and Representation Learning

The efficacy of any ML model is contingent upon the quality, quantity, and representativeness of the training data. The field has witnessed a paradigm shift from small, heterogeneous datasets extracted from literature to massive, standardized, and high-fidelity computational databases.

2.1.1 Large-Scale Computational Datasets The Open Catalyst Project (OC20) represents a landmark in data availability, providing over 1.28 million DFT relaxation trajectories covering diverse elemental compositions and adsorbates [4]. This dataset has enabled the training of deep graph neural networks (GNNs) that approach DFT accuracy for energy and force predictions, although challenges in out-of-domain generalization persist [3], [15]. Building on this, the OC22 dataset extended the scope to oxide electrocatalysts, introducing total energy targets and spin-polarization to capture the complex electronic structures of transition metal oxides [15], [16]. Most recently, the OC25 dataset addressed the critical limitation of vacuum conditions by generating 7.8 million DFT calculations for solid-liquid interfaces with explicit solvent and ions, enabling the training of models that account for solvation effects essential for realistic electrocatalysis [17].

Beyond the Open Catalyst series, specialized databases have emerged to target specific material classes. The BEAST DB provides grand-canonical DFT (GC-DFT) data computed at self-consistent fixed potentials, revealing significant deviations from the computational hydrogen electrode (CHE) approximation, particularly for chemical steps sensitive to bias [18]. Similarly, the CATALIUST dataset offers 60,000 DFT calculations specifically for Ti₂CTy MXenes with varied surface terminations, facilitating the development of machine-learned interatomic potentials (MLIPs) for dynamic surface chemistry [19]. These datasets collectively form the backbone for training foundation models and MLIPs that serve as surrogates for expensive quantum mechanical calculations.

2.1.2 Text Mining and Knowledge Extraction Complementing computational data, the vast corpus of scientific literature contains unstructured knowledge regarding synthesis protocols, experimental conditions, and performance metrics. Natural language processing (NLP) techniques

are increasingly employed to extract and structure this information. MatNexus, a comprehensive text mining suite, utilizes Word2Vec embeddings to capture latent semantic relationships between chemical elements and properties from millions of abstracts, enabling label-free similarity screening and composition embedding [20], [21]. Studies have shown that lightweight Word2Vec baselines can achieve superior candidate reduction compared to complex transformers when filtering combinatorial libraries, effectively mapping compositions onto descriptor spaces defined by concepts like ‘conductivity’ and ‘dielectric’ [21], [22].

Large Language Models (LLMs) fine-tuned on domain-specific corpora further enhance knowledge extraction. CataLM, pre-trained on over 12,000 electrocatalytic papers and instruction-tuned on expert-annotated QA pairs, demonstrates superior performance in extracting numerical entities and synthesizing knowledge compared to general-purpose LLMs [9]. Similarly, LLMs have been used to automatically construct multidimensional chemical search spaces by parsing unstructured literature for synthesis parameters, thereby expanding the exploration space beyond human intuition [23]. However, reliance on abstracts rather than full-text data remains a limitation, potentially missing detailed synthesis parameters and negative results [20], [24].

2.1.3 Geometric and Graph Representations Representing atomic structures for ML models has evolved from hand-crafted descriptors to end-to-end geometric deep learning. Graph Neural Networks (GNNs) treat atoms as nodes and interactions as edges, naturally respecting the permutation invariance and locality of atomic systems. Architectures like SchNet, DimeNet++, and GemNet have become standard baselines, with GemNet-OC achieving state-of-the-art performance on OC20 by incorporating angular information and efficient message passing [4], [25].

Innovations in graph construction continue to improve efficiency and accuracy. PhAST introduces task-specific modifications, such as removing redundant sub-surface atoms to reduce graph size and inference time by 4.5x without sacrificing predictive signal [25]. Voronoi tessellation has been proposed as a parameter-free alternative to radius-based graphs, encoding solid angles and contact types to better respect crystal symmetry, although it struggles with vacuum connections in slab models [26]. Furthermore, disconnected graph representations, where adsorbate and catalyst sub-graphs are processed separately before fusion, have shown that relaxed energy prediction retains significant capability even without explicit inter-component geometric edges, suggesting that global configuration features are partially encoded in component embeddings [27].

Multimodal approaches are also gaining traction. QE-Catalytic and CatalyticMLLM couple E(3)-equivariant geometric encoders with large language model backbones, aligning 3D atomic graphs with textual descriptors in a shared latent space. This enables bidirectional structure-property mapping and text-only property prediction with accuracy surpassing vanilla GNNs [28], [29]. CatBERTa takes a different approach, utilizing pretrained RoBERTa encoders to process textual representations of adsorbate-catalyst systems, bypassing the need for explicit coordinate inputs while achieving competitive accuracy through subgroup error cancellation [30].

2.2 Supervised Learning and Descriptor Engineering

Supervised learning remains the workhorse for predicting catalytic properties such as adsorption energies, overpotentials, and stability metrics. The field has progressed from linear models on physical descriptors to complex ensemble methods and symbolic regression for interpretable discovery.

2.2.1 Classical and Ensemble Regression Traditional machine learning algorithms, including Random Forest (RF), Support Vector Regression (SVR), and XGBoost, are widely used due to their robustness and ability to handle non-linear relationships with limited data. Stacking ensembles that combine tree-based learners (RF, XGBoost) with kernel-based meta-learners (SVR) have achieved R-squared values of 0.98 for HER and 0.94 for OER predictions on the Catalysis-hub database, identifying compositional melting temperatures and electronic configurations as dominant features [7].

Feature engineering remains critical for these models. In the design of transition metal-incorporated carbon quantum dots, a Bayesian Genetic Algorithm optimized categorical variables by integrating Naïve Bayes posterior probabilities, converging to optimal experimental conditions with fewer trials than conventional genetic algorithms [31]. For MXene catalysts, recursive feature elimination (RFE) combined with Random Forest Regression reduced 125 atomistic and statistical features to a compact set, achieving a mean absolute error (MAE) of 0.37 eV in predicting hydrogen adsorption free energy [32]. Similarly, statistical feature engineering expanded atomic descriptors from 28 to 60 variables for g-C₃N₄-supported single-atom catalysts, improving prediction accuracy from $R^2=0.56$ to 0.81 [33].

2.2.2 Symbolic Regression and Interpretable Descriptors While black-box models offer high accuracy, symbolic regression techniques like SISSO (Sure Independence Screening and Sparsifying Operator) provide physically interpretable analytical formulas. SISSO has been successfully applied to derive compact descriptors for single-atom catalysts on MXenes, identifying Bader charge squared as a primary predictor for OER activity, outperforming genetic programming approaches [34]. In the context of high-entropy alloys, SISSO broke linear Brønsted-Evans-Polanyi (BEP) and d-band center correlations by identifying non-linear descriptors based on pure host and guest properties, bypassing expensive alloy supercell calculations [35].

Multi-task extensions of symbolic regression address data inconsistency across diverse experimental sources. SCMT-SISSO enforces sign constraints on multi-task coefficients, resolving inconsistencies in perovskite OER datasets while distilling universal descriptors with a leave-one-out cross-validation RMSE corresponding to only 47 mV error in overpotential [36]. Furthermore, neural network-guided symbolic regression workflows use permutation importance from deep learning models to filter noise before symbolic search, enabling the discovery of compact formulas involving LUMO energy for perovskite catalysts [37]. These interpretable models are crucial for uncovering underlying physical mechanisms, such as the role of unpaired d-electrons in nitrogen reduction or the impact of lattice strain on binding energies [38], [39].

2.2.3 Physics-Informed and Theory-Infused Models Integrating physical laws directly into ML architectures enhances generalizability and data efficiency. Theory-infused Neural Networks (TinNet) embed the Newns-Anderson Hamiltonian as a differentiable layer, enforcing d-band chemisorption physics while learning from data. This approach outperforms fully connected networks and matches GNN accuracy on out-of-sample single-atom alloy systems [40]. Similarly, the wACSF-ML-Microkinetic framework utilizes weighted Atomic Center Symmetry Function descriptors that incorporate electronegativity and valence electrons, enabling universal screening across alloys, oxides, and single-atom catalysts for hydrogen peroxide synthesis [41].

Physics-informed machine learning also extends to kinetic modeling. The “Waveguide Kinetics” framework treats polarization curves as power-flow responses, using a theory-neutral modal representation to capture exponential regimes and saturation without presuming specific mechanistic the-

ories, achieving $<5\%$ RMSE across divergent datasets [42]. In another instance, a metal-dependent kinetic model replaced the universal rate constant in the Nørskov model with a term linearly dependent on the absolute hydrogen adsorption free energy, reducing prediction discrepancies by four orders of magnitude [43].

2.3 Machine-Learned Interatomic Potentials and Dynamics

Machine-learned interatomic potentials (MLIPs) bridge the gap between the accuracy of DFT and the speed of classical force fields, enabling long-timescale molecular dynamics (MD) simulations and extensive configurational sampling.

2.3.1 Equivariant Potentials and Foundation Models Modern MLIPs leverage E(3)-equivariant architectures to ensure rotational and translational invariance. Models like MACE, NequIP, and Allegro have demonstrated near-DFT accuracy for forces and energies. The RACE MLP, an equivariant GNN potential, achieved an MAE of 17 meV for binding energy prediction in dual-atom catalysts, enabling the systematic mapping of 20 distinct nitrogen coordination motifs [44]. Foundation models pre-trained on diverse datasets, such as MACE-MP-0, can be fine-tuned for specific tasks with minimal data. For instance, fine-tuning MACE on DFT-generated configurations of monolayer amorphous carbon revealed that structural disorder creates a broad distribution of adsorption sites, with seven-membered rings and surface curvature acting as key activity descriptors [45].

MLIPs are indispensable for studying dynamic phenomena inaccessible to static DFT. Simulations of high-temperature MOF pyrolysis using fine-tuned MACE models tracked copper-promoted linker decomposition and nanoparticle formation, revealing that foundational models exceed error thresholds without task-specific fine-tuning [46]. In the study of bimetallic nanoparticles, the FLARE-Allegro pipeline combined active learning with equivariant networks to model PdAu segregation under catalytic cycling, discovering that Pd atoms preferentially segregate into the bulk, reducing surface coverage and causing deactivation [6].

2.3.2 Accelerating Structure Relaxation and Screening MLIPs significantly accelerate structure relaxation and high-throughput screening. The Catlas framework orchestrates a two-stage coarse-to-fine inference pipeline, using pretrained GNNs to filter binary intermetallics before detailed microkinetic analysis, achieving near-DFT accuracy for $^*\text{CO}$ adsorption with MAE of 0.16 eV [47]. Direct generation of equilibrium adsorption structures via diffusion bridge models (DB-Cata) learns the transformation manifold between initial and relaxed states, reducing displacement errors by 3x compared to traditional methods [48].

Multi-fidelity screening funnels integrate MLIPs with DFT to maximize efficiency. In the discovery of high-entropy MBenes for CO_2 reduction, a workflow initiated with MCSQS generation, filtered by DFT formation energies, and then employed MACE MLIPs to accelerate phonon and adsorption screening across 1,375 configurations, identifying candidates with limiting potentials of 0.00 V [49]. Similarly, cloud-scale AI-HPC workflows utilized M3GNet potentials to screen 32.6 million candidates for solid-state electrolytes in under 80 hours, a task infeasible with pure DFT [8].

2.4 Generative Design and Inverse Materials Discovery

Generative models enable the inverse design of catalysts by proposing novel structures conditioned on desired properties, moving beyond screening existing databases.

2.4.1 Generative Flow Networks and Diffusion Models Generative Flow Networks (GFlowNets) offer a principled approach to sampling diverse catalyst structures proportionally to their performance metrics. The Catalyst GFlowNet sequentially constructs catalyst surfaces by selecting space groups, compositions, and lattice parameters, integrating M3GNet relaxation within the sampling loop to enforce stability. This method avoids convergence to a single optimum, instead generating a diverse set of stable structures for the hydrogen evolution reaction [50].

Diffusion models have also been adapted for crystal structure generation. MatterGen and related diffusion-based frameworks, when integrated with physical principle-based restrictions, have successfully identified low-energy inorganic electrides by restricting the search to electron-rich compositions [11]. For amorphous multi-element materials, the ApolloX framework combines conditional continuous diffusion variational autoencoders with particle swarm optimization, encoding chemical short-range order via pair density matrices to generate low-energy structures with 75.2% success rates [51]. In the realm of polymers, conditional generative models based on GPT and diffusion architectures have designed polymer electrolytes with ionic conductivities twice as high as the best training set candidates [52], [53].

2.4.2 Language Model-Based Generation Treating crystal structures as sequences of tokens allows the application of large language models for generative design. CatGPT, a GPT-2 based model, encodes lattice parameters and atomic coordinates into tokenized strings, employing a bypass mechanism to resolve atomic overlaps during decoding, achieving 99.7% structural validity [10]. Materials Transformers (MT) benchmarked various architectures on expanded element sequences, demonstrating that causal language models learn chemical grammars yielding 97.54% charge neutrality, significantly outperforming random sampling [54].

Retrieval-Augmented Generation (RAG) enhances LLMs by grounding their generation in physical constraints. A RAG-LLM framework integrated GPT-4 with a vector database of 50,000 materials, using two-stage retrieval to guide composition generation. This approach achieved 82% thermodynamic stability without fine-tuning, a 3.6x improvement over non-RAG baselines [55]. Furthermore, LacMaterial employs cross-domain analogy retrieval to inspire structural redundancy in solid-state electrolytes, transferring abstract relational principles from non-material domains to propose novel compositions validated by surrogate models [56].

2.4.3 Evolutionary and Reinforcement Learning Strategies Evolutionary algorithms and reinforcement learning (RL) provide alternative strategies for navigating the chemical space. The MAGECS framework integrates a Crystal Diffusion Variational Autoencoder with a Bird Swarm Algorithm to optimize latent vectors, generating alloy surfaces with a 2.5-fold increase in high-activity structures compared to training data [57]. In molecular optimization, MOSAiCS employs evolutionary Monte Carlo with graph-based crossover moves to traverse chemical space for electrolyte design, outperforming best molecules in the QM9 dataset [58].

Deep RL agents like AdsorbRL utilize goal-conditioned Deep Q-Networks to traverse composition space via element addition/removal actions. By constraining action spaces to valid subgraphs via Random Edge Traversal, the agent efficiently discovers compositions with improved adsorption energies, avoiding stagnation in invalid states [59]. Hierarchical generative modeling further decomposes design into coupled geometric and chemical sub-problems, using genetic algorithms for spatial arrangement and generative models for subunit chemistry, achieving a 30% reduction in activation barriers for Claisen rearrangement catalysts [60].

2.5 Autonomous Discovery and Self-Driving Laboratories

The ultimate realization of AI-driven discovery is the closed-loop autonomous laboratory, where AI plans experiments, robots execute them, and models update based on new data.

2.5.1 Active Learning and Bayesian Optimization Active learning (AL) strategies are central to minimizing experimental cost by selecting the most informative samples. The DASH workflow integrates data mining, active learning, and domain adaptation to discover acidic OER catalysts, iteratively refining quaternary oxide compositions and achieving an overpotential of 154 mV, surpassing 99.5% of literature samples [61]. Bayesian Optimization (BO) has been successfully applied to combinatorial sputtering libraries, identifying global activity optima in quaternary Ni-Pd-Pt-Ru systems after exploring only 20% of the composition space [62].

Advanced acquisition functions and surrogate models enhance AL efficiency. The LOCAL framework employs a hierarchical graph convolutional network with active learning to predict the stability of dual-atom catalysts, using embedding-space neighbor selection to target uncertain regions [63]. Convex hull-aware active learning (CAL) directly optimizes the entropy of the induced convex hull distribution rather than surrogate energy variance, converging on stable regions in 50 iterations compared to >150 for baseline methods [64]. Noise-aware active learning in autonomous scanning electrochemical cell microscopy (SECCM) embeds analytical voltammetry expressions into Gaussian processes, decoupling intrinsic kinetics from mass transport and accurately predicting trends after measuring only 15% of the library [65].

2.5.2 Multi-Agent Systems and LLM Orchestration Multi-agent LLM frameworks introduce reasoning and planning capabilities to autonomous discovery. MAESTRO orchestrates four specialized agents (Design, Reflect, Summary, Exploration) to design single-atom catalysts, breaking conventional scaling relations via in-context learning and achieving an ORR overpotential of 0.31 V, below the theoretical limit [12]. Catalyst-Agent utilizes a modular Model Context Protocol (MCP) architecture, where five specialized servers handle candidate seeding, structure retrieval, and energy evaluation, enabling end-to-end autonomous screening without monolithic codebases [66].

ChemNavigator employs a hierarchical multi-agent system with an Orchestrator, Scientist, and Designer to derive quantitative design rules for organic photocatalysts, autonomously recovering established chemical principles like resonance and induction through systematic computational experimentation [67]. These systems demonstrate that LLMs can effectively parse unstructured knowledge, generate hypotheses, and plan complex workflows, although they still face challenges in thermodynamic reasoning and synthesis feasibility assessment [66], [55].

2.5.3 Robotic Platforms and Closed-Loop Workflows Physical implementation of self-driving laboratories is advancing rapidly. The MAP-E platform integrates eight independently addressable electrochemical cells with robotic sample transfer, enabling parallel uncertainty-driven sampling and reducing standard deviation in pitting potential measurements by 4x compared to manual ASTM standards [14]. ALBATROSS automates the entire battery testing workflow, from electrolyte formulation to coin-cell assembly and evaluation, achieving reproducibility comparable to expert researchers (RSD < 1.2%) [68].

Closed-loop frameworks also extend to computational-experimental hybrids. The OCx24 pipeline integrated high-throughput synthesis of 572 samples with computational screening of 19,406 materi-

als, recovering a Sabatier volcano that correctly identified Pt as the apex catalyst without Pt in the training set [2]. In methane oxidation, an AI-assisted volcano-type screening workflow combined subgroup discovery with active learning to identify Ba₂GaTaO₆ as a superior catalyst, validated experimentally with a C₂-yield of 13.4% [69]. These platforms exemplify the synergy between high-throughput computation, robotic experimentation, and ML-guided decision making.

2.6 Stability, Degradation, and Operando Modeling

Predicting catalyst stability and understanding degradation mechanisms are as critical as activity prediction. ML is increasingly applied to model dynamic surface evolution and operando conditions.

2.6.1 Thermodynamic and Kinetic Stability Prediction Thermodynamic stability is often assessed via convex hull distances, but kinetic factors are equally important. The LOCAL framework predicts stability energies of dual-atom catalysts by first predicting integrated crystal orbital Hamilton population (ICOHP) values to mimic relaxed geometries, achieving an MAE of 0.145 eV [63]. Pourbaix diagram construction via DFT thermodynamic modeling reveals that metal centers in M@N₄-graphene SACs restructure under operating conditions, with metals having d-band filling $n > 8$ shifting reactivity to the carbon lattice [70].

Degradation mechanisms such as dissolution and phase transformation are being quantified. Atom probe tomography (APT) combined with ML analysis revealed that accelerated degradation in PdAu electrocatalysts triggers preferential Pd dissolution and surface Au segregation, alongside boron leaching [71]. Online SFC-ICP-MS tracking showed that Sn and Fe additives stabilize Pd against dissolution during ethanol oxidation, whereas Nb induces severe destabilization [72]. ML models trained on degradation data can distinguish between reversible operational blocking (e.g., bubble adsorption) and irreversible material degradation, advocating for standardized S-number quantification [5], [73].

2.6.2 Operando and Constant-Potential Simulations Modeling catalysts under operando conditions requires accounting for potential-dependent charge redistribution and solvation. Constant-potential DFT methods, such as ESM-RISM, reveal that Fe-N₄-C and Co-N₄-C SACs exhibit comparable ORR activity when potential-dependent charge states are considered, contrasting with constant-charge predictions [74]. The constant potential reactor framework integrates variable electronic neural network potentials with a modified Nosé-Hoover Lagrangian to maintain constant electrode potential during MD simulations, revealing surface reconstruction phenomena invisible to constant charge models [75].

Coverage-aware multiscale modeling reconceptualizes surface coverage as a Boltzmann-weighted ensemble of interconverting adsorbate motifs. In nitrate electroreduction, this approach identified that activation barriers collapse into a single linear relationship with excess charge on reacting Cu atoms, providing a unified descriptor for complex reaction networks [76]. Such advanced modeling techniques are essential for bridging the gap between static vacuum calculations and dynamic electrochemical environments.

3. Challenges and Outlook

Despite significant progress, several critical challenges hinder the widespread adoption of ML in electrocatalyst discovery. Addressing these issues is essential for transitioning from academic benchmarks to industrial impact.

3.1 Data Scarcity, Bias, and Quality

While large datasets like OC20 exist, they are predominantly composed of unary and binary systems, with limited representation of complex multicomponent alloys, oxides, and realistic surface terminations [15], [17]. Experimental data remains fragmented, noisy, and biased towards positive results, limiting the training of robust models for real-world conditions [77], [78]. The lack of standardized reporting protocols for activity and stability metrics further complicates data aggregation and model validation [73], [79]. Future efforts must prioritize the creation of FAIR (Findable, Accessible, Interoperable, Reusable) databases that include negative results, detailed synthesis parameters, and operando characterization data.

3.2 Generalizability and Out-of-Distribution Performance

ML models often struggle to generalize to out-of-domain (OOD) compositions and structures. GNNs trained on OC20 show significant performance drops when applied to oxides or materials with unseen elemental combinations [15], [27]. The “extrapolation problem” is particularly acute for generative models, which may propose thermodynamically unstable or synthetically inaccessible structures [11], [80]. Techniques such as domain adaptation, transfer learning, and physics-informed constraints are promising but require further development to ensure reliable predictions in unexplored chemical spaces [61], [40].

3.3 Accuracy vs. Computational Cost Trade-offs

There is an inherent trade-off between the accuracy of MLIPs and their computational cost. While foundation models like MACE offer high accuracy, their inference cost can be prohibitive for large-scale screening or long-timescale dynamics [49], [46]. Conversely, lighter models may lack the fidelity to capture subtle electronic effects crucial for catalysis [25], [81]. Developing hierarchical modeling frameworks that dynamically switch between low-fidelity surrogates and high-fidelity DFT based on uncertainty estimates is a promising direction to balance efficiency and accuracy [82], [83].

3.4 Uncertainty Quantification and Decision Making

Reliable uncertainty quantification (UQ) is critical for active learning and risk assessment in catalyst design. Many current models provide point predictions without confidence intervals, leading to overconfidence in erroneous predictions [84], [83]. Bayesian neural networks, ensemble methods, and conformal prediction techniques are being explored to provide calibrated uncertainty estimates, but their integration into standard workflows remains limited [85], [64]. Robust UQ is essential for guiding autonomous experiments and avoiding costly dead-ends in the discovery process.

3.5 Experimental Validation and Synthesis Feasibility

A significant gap exists between computationally predicted catalysts and their experimental realization. Many high-performing candidates proposed by ML models are difficult to synthesize due to kinetic barriers, phase instability, or complex precursor requirements [12], [86]. Integrating synthesis feasibility predictors and retrosynthesis planning tools into the discovery loop is crucial [87], [23]. Furthermore, the “materials gap” between idealized single-crystal models and real-world nanoparticles with defects, supports, and dynamic restructuring necessitates more realistic modeling approaches and rigorous experimental validation [6], [88].

3.6 Safety and Ethical Considerations

As AI systems become more autonomous, safety and ethical considerations emerge. Autonomous laboratories handling hazardous chemicals require robust safety protocols and fail-safes to prevent accidents [13]. Additionally, the reliance on proprietary datasets and models may exacerbate inequalities in access to advanced discovery tools. Promoting open-source frameworks, shared data repositories, and transparent reporting standards is essential for equitable progress in the field [77], [78].

4. Conclusion

Machine learning has fundamentally transformed the landscape of electrocatalyst discovery, offering unprecedented capabilities for high-throughput screening, inverse design, and autonomous experimentation. From the development of massive datasets like OC20 and OC25 to the deployment of multi-agent LLM systems and self-driving laboratories, the field has made remarkable strides in accelerating the identification of efficient and stable catalysts. Methodological advances in graph neural networks, machine-learned interatomic potentials, and generative models have enabled the exploration of vast chemical spaces with near-DFT accuracy.

However, significant challenges remain in ensuring model generalizability, quantifying uncertainty, and bridging the gap between theoretical prediction and experimental synthesis. Future research must focus on integrating physics-based constraints, improving data quality and standardization, and developing robust closed-loop frameworks that seamlessly combine computation, robotics, and AI reasoning. By addressing these challenges, the community can unlock the full potential of AI-driven discovery, paving the way for the rapid deployment of next-generation electrocatalysts essential for a sustainable energy future. The convergence of advanced algorithms, high-performance computing, and automated experimentation promises a new era of materials science where the discovery cycle is measured in days rather than decades.

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